National University of Computer and Emerging Sciences

Lab Exercise 09
AL2002-Artifical Intelligence Lab

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Exercise

1. Load the given dataset and perform data cleaning (*you may use your code from Lab 8.*)

2. **Regression: Predicting Life expectancy**
    a. Separate the dataset into Features (X) and target label (y)
    b. Split the dataset into training and testing sets using: test_size = 0.2 and random_state = 42
    c. Apply appropriate feature scaling and label encoding techniques where required.
    d. Train a Linear Regression model from scikit-learn
    e. Evaluate the performance of your test dataset on this model using the following evaluation criteria:
        - Mean Squared Error (MSE)
        - Root Mean Squared Error (RMSE)
        - R^2 Score
    f. Train another regression model from scikit-learn (you can explore other models here: https://scikit-learn.org/stable/supervised_learning.html)
    g. Compare the performance of both models on test set and identify which one performs better.

3. **Classification: Predicting Country Status**
    a. Separate the dataset into Features (X) and target label (y)
    b. Split the dataset into training and testing sets using: train_size = 0.7 and random_state = 42
    c. Apply appropriate feature scaling and label encoding techniques where required.
    d. Train two different classification models from scikit-learn of your choice (you can explore a few models here: https://scikit-learn.org/stable/supervised_learning.html)
    e. Evaluate the performance of your test dataset on these models using the following evaluation criteria:
        - Accuracy Score
        - Confusion Matrix
        - Classification Report
    f. Compare the performance of both models on test set and identify which one performs better.